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Project Title:	The Relation Between Cognitive Injury, Network Stability, and Epigenetic Change Following Exposure to Space Radiation		
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Space Biology Special Category:	None		
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Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
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COI Name (Institution):	Impey, Soren (Oregon Health & Science University) Turker, Mitchell (Oregon Health & Science University) Rosi, Susanna (University of California San Francisco)		
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Task Description:

Space radiation poses a significant hazard to astronaut health both during the mission and at later times. However, uncertainty exists with respect to the level of risk to the CNS (central nervous system) following radiation doses relevant to those encountered during space missions. Radiation exposure in the brain affects the hippocampus, a structure critical for cognition. This proposal addresses section 1.2 of NASA Research Announcement NNJ12ZSA001N entitled, “Risk of Acute or Late Central Nervous System Effects from Space Radiation.” The proposed work focuses on changes in hippocampal network stability, including epigenetic mechanisms, in the development of cognitive dysfunction after space irradiation. Radiation exposure of the CNS significantly affects the hippocampus. Our preliminary data show impaired cognition in mice at two weeks following low dose space radiation. During spatial exploration, when an animal enters a specific location to which the cell responds (known as the cell's place field), hippocampal pyramidal neurons in the CA areas display robust activity. The place fields create an environment-specific "map," thought to be critical in establishing the context of a memory and resulting in the transcription of the immediate early gene (IEG) activity-regulated cytoskeletal associated protein (Arc). Thus, analyzing Arc provides a reliable method to detect the number of active cells forming environment-specific maps and to capture the environmental specificity of place fields. That is, place fields are highly reproducible and stable across multiple visits to the same environment. In contrast, this stability is not seen across visits in different environments. The proposed work will focus initially on effects of space radiation on the stability of hippocampal networks, assessed by analyzing the expression of Arc and other genes important for memory formation. We hypothesize that the detrimental effects of space radiation on hippocampus-dependent cognition are associated with changes in hippocampal network stability. Promoter region DNA methylation (5mC) of Arc leads to decreased expression and space radiation induces changes in DNA methylation. Moreover, DNA methylation plays a key role in regulating expression of genes required for spatial learning and memory. Thus, study of Arc expression and DNA methylation after charged particle exposure is warranted, followed by a more global search for DNA methylation changes in genes involved in learning and memory. A second form of DNA methylation, termed 5-hyrdroxymethylation (5hmC), is abundant in the brain and found at promoters of active genes, unlike 5mC, which is found at the promoters of inactive genes. In this proposal, we will test the hypotheses that the detrimental effects of space radiation on cognition are associated with Arc-related changes in the hippocampal networks involved in learning and memory and that these changes are related to space radiation-induced perturbation of 5mC and/or 5hmC. Our approach will link assessment of hippocampus-dependent cognitive changes, hippocampal network stability, and alterations in global DNA methylation and expression of genes involved in learning and memory. In Specific Aim 1, we will test the hypothesis that low-dose whole body irradiation with 56Fe or 28Si ions or protons has short- and long-term effects on cognitive performance in object recognition and spatial learning and memory requiring navigation. In Specific Aim 2, we will test the hypothesis that low dose whole body irradiation with 56Fe or 28Si ions or protons has short- and long-term effects on hippocampal networks involving Arc, and that cognitive injury following space irradiation is associated with changes in hippocampal network stability. In Specific Aim 3, we will test the hypothesis that cognitive changes following space irradiation are associated with alterations in DNA methylation and expression of Arc and other genes involved in learning and memory.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Humans are exposed to ionizing irradiation under a variety of conditions associated with medical treatment. While the risks of developing cognitive impairments in those situations are fairly well established, uncertainty exists with respect to other types of radiation exposure, such as those experienced in the space environment. There is a critical need to establish the level of risk to the CNS following radiation doses relevant to those encountered during space missions. NASA has identified four categories of radiation risk in the Integrated Research Plan (IRP) and their associated research and technology gaps. This project is pertinent to 1.2 “Risk of Acute or Late Central Nervous System Effects from Space Radiation” and to Space Radiation Risks CNS1, CNS2, CNS6, Degenerative 1, and Degenerative 2. The project focuses on the role of changes in hippocampal network stability and epigenetic mechanisms involved in these changes in the development of cognitive dysfunction after space irradiation. CNS-related topics expressly targeted by the IRP and that are specifically or indirectly addressed in the current project include early and late effects of space irradiation on cognitive function and cellular and molecular biomarkers that might be associated with cognitive injury.
Innovative technologies are being developed for integrated functional assays to assess effects of space radiation on brain function.

Task Progress:	This project focuses on changes in hippocampal network stability including epigenetic mechanisms in the development of cognitive dysfunction after space irradiation. Radiation exposure of the central nervous system (CNS) can significantly affect the hippocampus, a structure critical for cognitive function. During spatial exploration, when an animal enters a specific location a distinct cell responds (known as the cell's place field). As a result of these place fields, hippocampal pyramidal neurons in the CA areas display robust high frequency activity. The place fields create an environment-specific "map," thought to be critical in establishing the context of a memory. Following visits to an environment for a second time, the same cells form stable neural networks and respond as a result of cellular activity induced by a learning experience. While place fields are highly reproducible and stable across multiple visits to the same environment, this stability is not seen in when different environments are visited. The detrimental effects of charged particles on hippocampus-dependent cognition are likely associated with changes in hippocampal networks. Analysis of the precise temporal kinetics of the plasticity-related immediate early gene (IEG) activity-regulated cytoskeletal associated protein (Arc) using a method called catFISH (cellular compartment analysis of temporal activity using fluorescence in situ hybridization) provides an ideal and reliable approach for studying network stability. In a prior study we showed that Arc protein expression in the hippocampus is sensitive to brain-only exposure to 56Fe ions (600 MeV, 1 Gy). Arc expression is reduced with aging in the CA1 “place cells” within the mouse hippocampus and this reduction is correlated with increased promoter region DNA methylation (5mC). This is not surprising because changes in DNA methylation play a key role in regulating expression of genes required for spatial learning and memory. Gamma and space irradiation, including 56Fe ions and protons, induce changes in DNA methylation. Of significance for this project, recent studies show that a second form of DNA methylation, termed 5-hyrdroxymethylation (5hmC) is abundant and stable in the brain. The TET (ten eleven translocation) enzymes convert 5mC to 5hmC, which likely play a role in gene regulation because 5hmC is often found at promoters of active genes. In contrast, 5mC is found at the promoters of inactive genes. Changes in 5mC, which are induced by environmental exposures including radiation and behavioral modification, play a role in processes including cancer and neurological disease. A similar role for 5hMC can be assumed due to its abundance in the brain, though 5hmC changes after radiation exposure and in neurological disease has not yet been elucidated. We hypothesize that the detrimental effects of charged particles on hippocampus-dependent cognition are associated with Arc-related changes in the hippocampal networks involved in synaptic plasticity and memory. We further hypothesize that these changes are related to perturbation of 5mC and/or 5hMC in the hippocampus that affect Arc expression and expression of other learning and memory genes in the same signaling pathways and networks. In Yr 1, we finalized the analysis of the novel object recognition data of mice irradiated at BNL and tested 2 weeks after irradiation. The mice were all 6-month-old male C57Bl6/J mice (no retired breeders). The doses of irradiation were: 56Fe (600 MeV/n; 0.1, 0.2, and 0.4 Gy); 28Si (600 MeV/n; 0.3, 0.6, and 0.9 Gy); and protons (150 MeV, 1 Gy). For the novel object recognition test, the mice were habituated to an open field for three subsequent days. Strikingly, the pattern of radiation effects on novel object recognition was identical to that seen for Arc in a subregion of the hippocampus. We determine if the neurons responding to an initial experience (exploration of environment A) are the same neurons that respond to a second and identical experience as expected based on sham-irradiated animals (AA Paradigm), and if the hippocampal networks activated during exploration of the first environment are statistically independent from those activated by the exploration of a different environment (AB Paradigm). With regard to the epigenetic analysis in Yr 1 of the grant, there was a bias towards upregulation of 5hmC. This may reflect the relative stability of this mark in brain (~20-30% of modified cytosines in hippocampus are 5hmC). Differentially regulated regions were selected based on their proximity to transcriptional start sites (25 kb window) and preliminary gene ontology and pathway analyses were conducted. Interestingly, only upregulated regions were significantly associated with gene categories. The radiation dose was particularly enriched for gene categories and pathways associated with neuronal maturation, synapse remodeling and cell adhesion. We also observed enrichment for signaling pathways-associated with neuronal remodelling (Wnt signaling) and synaptic plasticity (MAPK signaling). These data suggest that exposure to space radiation triggers long-term epigenetic changes associated with genes that regulate structural changes in the brain.
Bibliography Type:	Description: (Last Updated: 10/30/2023)
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